

81-LED CHASER LIGHT Using CD4017

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This is a simple 81-LED chaser light using two CD4017 decade counters. NE555 timer is used for clock generation for CD4017. We have used only two CD4017 ICs to construct the project in 9x9 matrix format. The circuit is useful for applications like running light systems in halls, decorations, dance programmes, counting and so on. Fig. 1 shows the author's prototype.

Circuit and working

Circuit diagram of the 81-LED chaser light using CD4017 is shown in Fig. 2. It has two CD4017 decade counters

PARTS LIST

Semiconductors:

- IC1 - 7805 5V regulator
- IC2 - NE555 timer
- IC3, IC4 - CD4017 decade counter
- T1-T18 - BC547 npn transistor
- LED1-LED81 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R9 - 220-ohm
- R10-R18 - 10-kilo-ohm
- R19 - 4.7-kilo-ohm
- R20 - 1-kilo-ohm
- VR1 - 5-kilo-ohm potmeter

Capacitors:

- C1 - 10 μ F, 16V electrolytic
- C2-C4 - 0.01 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- 9V battery

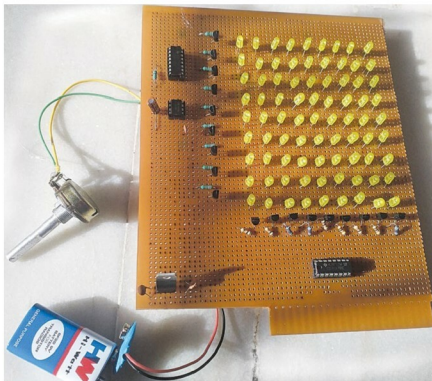


Fig. 1: Author's prototype

(IC3 and IC4) and an NE555 timer (IC2). The 9V battery power supply is stepped down to 5V using a 7805 voltage regulator along with two 0.1 μ F capacitors (C3 and C4) to eliminate any ripples in supply voltage.

IC2 is wired in astable multivibra-

tor mode. For clock frequency generation, 10 μ F capacitor (C1), resistor R20 and potmeter VR1 are used. VR1 is used to change the LEDs' flashing rate. Frequency (f) can be calculated using the relationship:

$$f = 1 / (0.69(R20 + 2 \times VR1) \times C1)$$

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